

Introduction to Cosmology and Dark Energy

Workshop on Astrophysics and Cosmology 2026
SXPC-NEPAL

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February 4, 2026

- Basics of cosmology
- Dark Energy
- Λ CDM model
- Dynamical Stability Analysis
- MCMC fitting with Hubble data.

Overview of Cosmology

The Big Bang Theory explains the origin and evolution of the universe.....

Overview of Cosmology II

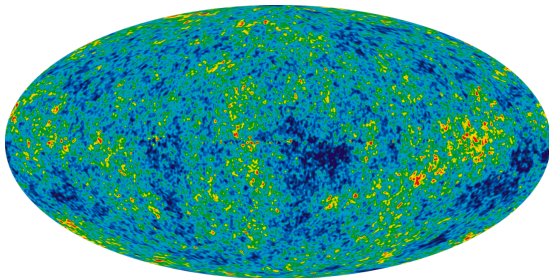


Figure: Cosmic Microwave Background Radiation, Image: WMAP/NASA

Cosmological Principle

According to the cosmological principle, the Universe is homogeneous and isotropic at large scales

- By homogenous, we mean that the distribution of matter and energy is uniform throughout the cosmos when averaged over very large scales
- By isotropic, it means that the universe looks the same no matter which direction one looks at, or simply we are not in a special place of the universe.

The Robertson-Walker Metric

The FRW Metric in a 4D manifold is given by

$$ds^2 = -dt^2 + a(t)^2 \left[\frac{dr^2}{1 - \kappa r^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right] \quad (1)$$

where $\kappa = (-1, 0, +1)$ corresponds to the open, flat, and closed universe, while $a(t)$ is a scale factor.

Dark Energy

The nature of dark energy is a pivotal question in cosmology, with implications for fundamental physics and the universe's fate.

The discovery of the universe's accelerated expansion in the late 1990s, through observations of Type Ia supernovae [5, 3], introduced dark energy as a dominant component driving this phenomenon.

Dark energy constitutes approximately 68% of the universe's energy in the Λ CDM model, and exerts negative pressure, counteracting gravitational collapse and accelerating cosmic expansion [4].

Despite its critical role, the physical nature of dark energy remains one of cosmology's greatest mysteries, with candidates ranging from a cosmological constant to dynamical scalar fields [1].

The Λ CDM Model

This model is older than many modern astrophysicists realize, and it has not fundamentally changed for over twenty-five years [6].

Einstein's Equation

The dynamics of metric is given by Einstein's field equation $g_{\mu\nu}$:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa^2 T_{\mu\nu} \quad (2)$$

where R = Ricci scalar. For perfect fluid on large scales, the energy-momentum tensor can be written as,

$$T_{\mu\nu} = pg_{\mu\nu} + (\rho + p)u^\mu u^\nu \quad (3)$$

The relation between the energy density and pressure is stated by the equation of state

$$p = \omega\rho \quad (4)$$

($\omega = 0$ dust; $\omega = 1/3$ radiation).

Cosmological Equations

After solving the Einstein Field Equations 2 by the use of the FRW metric, the solution arises in two coupled differential forms for $a(t)$ and matter variables $\rho(t)$ and $p(t)$.

$$\frac{\kappa}{a^2} + H^2 = \frac{\kappa^2}{3}\rho \quad (5)$$

$$\frac{\kappa}{a^2} + 2\dot{H} + 3H^2 = -\kappa^2 p \quad (6)$$

Equation 5 is also called the Friedmann Equation, and Equation 6 is called the acceleration equation. Using the above equations,

$$\frac{\ddot{a}}{a} = -\frac{\kappa^2}{6}(\rho + 3p) \quad (7)$$

Acceleration if $\rho + 3p < 0$.

Energy Conservation Equations

The Friedmann and acceleration equation gives the continuity equation,

$$\dot{\rho} + 3H(\rho + p) = 0 \quad (8)$$

Then, using the linear equation of state, this equation gives the solution,

$$\rho \propto a^{-3(\omega+1)} \quad (9)$$

For Dust/Radiation

$$\rho_m \propto a^{-3}, \quad \rho_r \propto a^{-4} \quad (10)$$

Cosmological Constant

Now, let's have a look at the dark energy model driven by the cosmological constant. Λ was first introduced by Einstein in his field Equations 2, and can be rewritten as,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu} \quad (11)$$

Then the new cosmological equations becomes,

$$3H^2 = \kappa^2 \rho + \Lambda \quad (12)$$

$$2\dot{H} + 3H^2 = -\kappa^2 p + \Lambda \quad (13)$$

Now, since $\omega_\Lambda = -1$ ($p_\Lambda = -\rho_\Lambda$). Consider the cosmological constants with both matter and radiation for which the values, ($p_m = 0$) and ($p_r = \rho_r/3$), then cosmological equations take the form,

$$3H^2 = \kappa^2 \rho_m + \kappa^2 \rho_r + \Lambda \quad (14)$$

$$2\dot{H} + 3H^2 = -\frac{\kappa^2}{3}\rho_r + \Lambda \quad (15)$$

Dynamical System and Physical Phase Space

Let us define dimensionless variables

$$x = \Omega_m = \frac{\kappa^2 \rho_m}{3H^2}, \quad y = \Omega_r = \frac{\kappa^2 \rho_r}{3H^2} \quad (16)$$

The Friedmann equation can be rewritten as:

$$1 = x + y + \Omega_\Lambda, \quad \Omega_\Lambda = \frac{\Lambda}{3H^2} \quad (17)$$

Dynamical Equations

To obtain a dynamical system, we differentiate variables x and y with respect to $\log a$.

$$x' = x(3x + 4y - 3) \quad (18)$$

$$y' = y(3x + 4y - 4) \quad (19)$$

Critical Points and Effective EoS

Now, we can obtain critical points of the system by equating dynamical equations to zero, $x' = 0$ and $y' = 0$, we get two points $x = 0$ and $y = 0$.
Now

$$\text{For, } y = 0 \quad x' = 0 \quad \implies 3x + 4 \cdot 0 - 3 = 0 \quad \implies x = 1 \quad (20)$$

$$\text{For, } x = 0 \quad y' = 0 \quad \implies 3 \cdot 0 + 4y - 4 = 0 \quad \implies y = 1 \quad (21)$$

This will lead to critical points $O(0,0)$, $R(0,1)$, and $M(1,0)$, and after performing stability analysis, we obtain the eigenvalues as shown in Table 1. The effective EoS parameter w_{eff} is given by

$$\begin{aligned} \omega_{eff} &= \omega_m \Omega_m + \omega_R \Omega_R + \omega_\Lambda \Omega_\Lambda \\ &= 0 \cdot \Omega_m + \frac{1}{3} \cdot \Omega_R + (-1) \cdot \Omega_\Lambda \\ &= -1 + x + 4y/3 \end{aligned}$$

Stability Analysis

Point	x	y	ω_{eff}	Eigenvalues	Stability
O	0	0	-1	$(-4,-3)$	Stable
R	0	1	1/3	$(1,4)$	Unstable
M	1	0	0	$(-1,3)$	Saddle

Table: Critical points and eigenvalues of the system for Λ CDM Model

Plotting Phase Space Diagram of Λ CDM Model

Phase Space Diagram

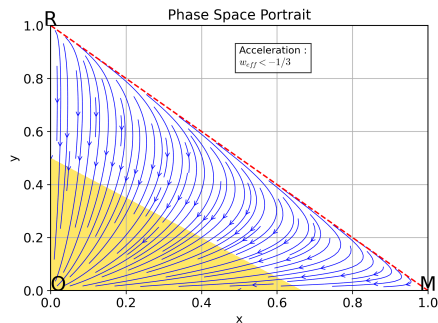


Figure: Phase Space Portrait of Λ CDM Model

- Trajectories: From radiation (R) to matter (M) to DE (O).
- Shaded: Accelerated expansion ($w_{eff} < -1/3$).

Phase Space Diagram

For scalar field cosmological models, we apply the following steps

- Action
- Find Equation of Motion, or Klein-Gordon Equation (From Euler-Lagrange Equation)
- Vary the action with respect to the metric which gives the Energy-Momentum Tensor
- Find Cosmological Equations
- Dimensionless Variables
- Eigenvalues and stability analysis

Open the `phantom` file in Google Collab. Use the autonomous equations for the phantom field and define a system. Obtain a phase space portrait.

$$x' = \frac{1}{2} \left[-3x^3 - 3x(1 + y^2) - \sqrt{6}\lambda y^2 \right] \quad (22)$$

$$y' = -\frac{1}{2}y \left[3x^2 + 3(y^2 - 1) + \sqrt{6}\lambda x \right] \quad (23)$$

Detailed Derivation

Visit [here](#) for more detailed derivation.

Phantom Phase Space

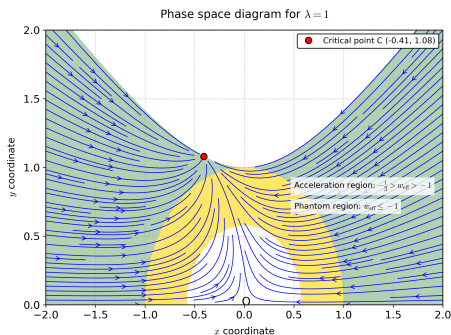


Figure: Phase space portrait of phantom dark energy model. The yellow shaded region indicates the acceleration region, and the blue shaded region indicates the phantom field-dominated region. Reference: [Ch 6, Phantom Fields](#)

MCMC Implementation Using Emcee

In this project, we use a popular Python package that uses the stretch-move algorithm, called `emcee`. `emcee` is a Python implementation of the affine-invariant ensemble sampler for Markov chain Monte Carlo (MCMC) proposed by Goodman & Weare [2]

Constraining DE Parameters using Hubble Data

Results

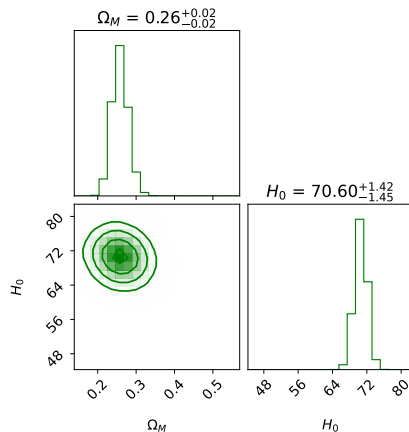


Figure: Corner Plot of Cosmological Constraints from Hubble Data

Conclusion and Need for Scalar Field Models

The standard model, while remarkably successful in explaining a broad array of cosmological observations such as the cosmic microwave background (CMB) anisotropies, large-scale structure formation, and the accelerated expansion of the universe, faces mounting challenges from persistent observational tensions that suggest the need for extensions or alternatives, particularly those involving scalar field dynamics to model dark energy more flexibly.

Additional Resources

Visit [archive](#) for lecture notes. All the resources used in these notes are available on the author's [GitHub Repository](#). For more cosmology codes and dynamical stability analysis of scalar field cosmological models, please visit another repository [here](#). If you have any queries or suggestions, you can contact the author directly via multiple [channels](#).

References



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